

DATA
GOVERNANCE
IN AFRICA



AI 4 STARTUPS

HACKATHON BEST PRACTICES

Implementation period – Final report

This report compiles best practices for organising hackathons for startups, drawn from the AI 4 Startups (AI4SU) program experience: 8 hackathons, more than 4 700 participants, 24 open-source AI models. It is intended as a practical guide for program managers, donors, and ecosystem actors seeking to replicate or adapt the model.

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EXECUTIVE SUMMARY

The AI 4 Startups (AI4SU) program – implemented by Digital Africa in collaboration with Expertise France under the European Data Governance in Africa Initiative – implemented hackathons as practical instruments for unlocking data sharing, building AI capacity, and delivering usable open-source models to African agritech startups.

Between April 2025 and March 2026, the program organised 8 hackathons across the AI4SU focus markets, engaging more than 4,700 participants and producing 24 open-source AI models – all published on the [AI4SU platform](#).

The program operated through two complementary modalities:

- in-person event hackathons (2–3 days, 50–130 participants, with local implementation partners in Accra and Nairobi)
- and online data challenges (4–8 weeks, 700–1 100 participants, hosted on the Zindi platform).

Each challenge was co-designed with a partner startup around its own operational data – from credit scoring for smallholder farmers to satellite-based crop classification – making the startup both the data provider and the direct beneficiary of the outputs.

This report distils the operational experience of the AI4SU program into a practical guide structured in five parts:

1. **Part 1** introduces the program, its institutional context, and the methodology used to compile these best practices.
2. **Part 2** defines what a hackathon is – and what it is not – through a typology of competition-based innovation formats, and positions the AI4SU model within the African startup ecosystem context.
3. **Part 3** details the AI4SU methodology: the dual-modality framework, three co-design pathways tested with different implementation partners, and the full operational cycle from startup identification to model diffusion – including a two-step Data Sharing Agreement (DSA) framework designed to protect startups without slowing innovation.
4. **Part 4** presents 9 best practices organised along three phases – design & preparation, execution, and diffusion & impact – covering startup screening, challenge co-design, data preparation, mentorship, participant recruitment, solution documentation, model delivery, and IP management.
5. **Part 5** synthesises five key learnings and identifies the conditions for replicating the model beyond agritech.

Five key learnings emerge

1. **Startups are the right actors – but fragile partners.** Their value proposition must be clear and tangible; the moment the process feels extractive, they disengage.

2. **Data quality determines everything downstream.** Every shortcut in data preparation produces unusable models.
3. **Data sharing requires staged trust, not just legal frameworks.** The two-step DSA approach — the program's most transferable contribution — builds confidence incrementally.
4. **Match the format to the objective.** If the goal is not a working output, it is not a hackathon — and calling it one creates unrealistic expectations.
5. **Hackathons are ecosystem infrastructure.** The talent surfaced, communities built, and open-source assets created have systemic value that exceeds any single competition's output.

The methodology is sector-transferable: the co-design cycle, data maturity framework, and DSA architecture can be adapted to healthtech, fintech, climate, logistics, or any domain where startups hold underexploited data assets. Four conditions are necessary: access to a competition platform, data science capacity in the preparation team, a legal framework for data sharing, and local implementation partners for in-person events.

I. INTRODUCTION

A. The AI4SU Program & the Data Governance in Africa Initiative

The Data Governance in Africa Initiative supports the African Union and its member states to form development-oriented and human-centric data regulation, data use, and data infrastructure at continental, regional, and member state levels. The Initiative provides concrete support for a single digital market in Africa by harmonising regulations on data, creating cross-border data flows, and promoting investments in secure and sustainable data infrastructure facilities. It is funded by €60 million from the European Union and five of its Member States (Belgium, Estonia, Finland, France, and Germany) under the Global Gateway strategy, and implemented from 2023 to 2026 by GIZ, Enabel, and Expertise France as implementing partners, and Digital Africa, ESTDEV, and HAUS as grantees.

Within this initiative, the AI 4 Startups (AI4SU) program – implemented by Digital Africa in collaboration with Expertise France – takes a startup-centred approach to data governance and AI development with and for Agritech startups. Rather than working top-down on regulatory frameworks, the program starts from the concrete data challenges that African startups face when building AI-driven products. Its objective: unlock the value of data sharing with and for startups, through structured hackathons, open-source model development, and applied data governance frameworks. The program developed a dual hackathon model – in-person event hackathons (2–3 days, hosted locally with startup partners) and online data challenges (4–8 weeks, hosted on the Zindi platform and open to participants continent-wide) – designed to combine the depth of face-to-face co-creation with the scale and benchmarking power of open online competitions.

Over the course of the program, AI4SU engaged more than sixty startups and organised around thirty workshops to identify, co-design, and refine data challenges suited to hackathon-based AI development. This process culminated in eight hackathons with agritech startups across multiple African markets, each structured around a specific data challenge. The hackathons produced 24 open-source AI models, now available to the broader ecosystem on the AI4SU platform.

#	Challenge	Country	Date	Modality	Startup Partner	Participants
1	Pixel-Level Crop Classification for Precision Agriculture	Côte d'Ivoire	Apr 2025	Online	Tolbi	866
2	Customer Segmentation & Retention Intelligence for Agri-Ecommerce	Ghana	Oct 2025	Event	AfriMash	50
3	Inclusive Credit Scoring for Smallholder Farmers	Kenya	Oct 2025	Event	FarmSawa	91

4	Maize Price Intelligence for Smarter Selling	Kenya	Nov 2025	Online	agriBORA	1 095
5	Produce Recommendation Systems for Food Waste Reduction	Kenya	Nov 2025	Online	Farm to Feed	733
6	Climate & Market Resilience	Kenya	Dec 2025	Event	FarmSawa	126
7	Predicting Practice Adoption in Smallholder Farming	Kenya	Jan 2026	Online	DigiCow	896
8	Soil Nutrient Prediction for Sustainable Farming	Kenya	Feb 2026	Online	Rhea	896

Table 1 – Overview of the 8 AI4SU hackathons (3 in-person events, 5 online data challenges). Total: +4 700 participants, 24 open-source AI models.

All program resources – open-source models, datasets, documentation, and partner publications – are accessible on the AI4SU platform (ai4su.digital-africa.co), which serves as the central knowledge hub for the program.

B. Purpose & Audience of This Report

This report compiles best practices for organising hackathons with and for startups, drawn from the AI4SU program experience. It is intended as a practical guide for:

- Program managers designing innovation competitions or data-sharing initiatives with startups
- Development finance institutions (DFIs) and donors seeking to integrate hackathon-based approaches into their support programs
- Innovation hubs and startup support organisations looking to replicate or adapt the model in their ecosystem

C. Methodology & Sources

This report draws on the following sources:

- 8 hackathons conducted between April 2025 and March 2026 across the AI4SU program scope, in two modalities (3 in-person events, 5 online data challenges hosted on the Zindi platform)
- 30+ co-design workshops and semi-structured interviews with partner startups, conducted during the challenge identification and preparation phase
- The Briter Intelligence ecosystem mapping (2026), covering 250+ AgTechs, 250+ funders, and 70+ data policies across 8 African markets
- The Cage Patrick & Company data and regulatory landscape study (2026), providing legal analysis of data-sharing frameworks across the program's focus countries

- The companion *AI & Data in African Agriculture Use Cases* report (Digital Africa, 2026), analysing startup data challenges and AI applications
- Digital Africa's operational expertise gained through the design and management of the AI4SU program: the co-design methodology, the structuring of data-sharing agreements, and the practical lessons learned from running hackathons as applied data governance experiments

II. HACKATHONS: CONCEPTS & TYPOLOGY

A. What Is a Hackathon?

The word *hackathon* is a blend of the words *hack* – in the sense of exploratory, creative programming – and *marathon*, referring to the sustained effort required from participants. The term first appeared in 1999, when two events independently coined it: a gathering of OpenBSD developers in Calgary, Canada, working on cryptographic systems, and a challenge organised by Sun Microsystems at its JavaOne conference, where participants were tasked with programming a new feature for the Palm V handheld device.

In its original and strictest sense, a hackathon is a **time-bounded, collaborative event** in which participants – typically developers, data scientists, designers, and domain experts – work intensively to produce a functional prototype or solution in response to a **defined challenge**. The requirement to produce a working, demonstrable output – not just ideas or recommendations – is what sets hackathons apart from other participatory innovation formats.

Over the following decade, the model spread rapidly beyond software engineering. Companies such as Facebook famously used internal hackathons to develop key features – including the "Like" button, the chat function, and the news feed (Keyani, P., "Stay Focused and Keep Hacking", Facebook Engineering Blog). By the 2010s, governments, NGOs, and international organisations had adopted the format for civic and social-impact purposes, giving rise to "civic hackathons" and "data-for-good" competitions.

Four criteria distinguish a genuine hackathon from other innovation events:

- **Time pressure** – a short, fixed duration (typically 24–72 hours in person, or a few weeks online) that forces rapid decision-making and prototyping
- **A defined challenge** – participants respond to a specific problem statement, not an open-ended brainstorm
- **A working output** – teams must produce a functional prototype, model, or demonstrable solution – not a slide deck, a set of recommendations, or a business plan.

- **Competition and peer evaluation** – a jury or leaderboard mechanism selects winning contributions

These criteria matter because the term "hackathon" is increasingly used loosely to describe a wide range of in-person events involving data or technology – many of which lack one or more of these defining features (see 2.3 – *Not a Hackathon, But a...* below).

B. A Typology of Competition-Based Innovation Formats

The landscape of competition-based innovation formats has diversified considerably since 1999. The table below maps the main models along key dimensions: duration, format, participant profile, typical organisers, expected outputs, and trade-offs.

Format	Duration	Setting	Target participants	Typical organisers	Primary output	Strengths	Limits
Coding Hackathon	24–72 h	In-person	Developers, engineers	Tech companies, universities	Working software prototype / MVP	Speed, team cohesion, talent scouting	Outputs often unfinished; high participant fatigue
Datathon / Online Data Challenge	1–8 weeks	Online (platform-based)	Data scientists, ML engineers, statisticians	Data platforms (Zindi, Kaggle, DrivenData), research orgs, development programs	Trained ML models, notebooks, benchmarks	Scale (hundreds to thousands of participants globally); reproducible, ranked outputs	No face-to-face interaction; limited ecosystem-building; requires well-prepared datasets
Créathon / Design Sprint	1–3 days	In-person	Designers, entrepreneurs, domain experts, students	Innovation hubs, incubators, corporates	Concept, user journey, low-fidelity prototype	Accessible to non-technical profiles; strong ideation	No functional output; risk of shallow solutions

Startup Weekend (Techstars)	54 h	In-person	Aspiring entrepreneurs, developers, designers	Techstars, local startup communities	Business model + pitch + basic prototype	Entrepreneurial learning; team formation; low barrier to entry	Business focus over technical depth; outputs rarely survive beyond the event
Innovation Prize / Grand Challenge	Months	Hybrid (online application + in-person finals)	Startups, research teams, social enterprises	DFIs, foundations, governments (e.g. XPRIZE, Grand Challenges)	Detailed business case, pilot project, proven concept	Deep solutions; significant funding; visibility	Long timelines; high selection barriers; fewer participants
Open Innovation Contest	Variable (weeks to months)	Hybrid or online	Broad (corporates, startups, individuals)	Corporates, industry consortia	Partnerships, IP, co-development agreements	Ecosystem engagement; access to external talent and ideas	IP ownership often unclear; integration of outputs into operations is slow

Table 2 – Typology of competition-based innovation formats. The AI4SU program used two formats: in-person event hackathons (combining coding hackathon and datathon elements) and online data challenges on the Zindi platform (datathon format).

The key differentiators across these formats are: **time horizon** (hours vs. weeks vs. months), **openness of data and code** (open-source vs. proprietary), **IP ownership** (participants vs. organisers vs. shared), and **participant profile** (technical specialists vs. mixed teams vs. entrepreneurs).

C. Not a Hackathon, But a...

The growing popularity of hackathons has led to a proliferation of events labelled as such without meeting the core criteria outlined in section 2.1. This is particularly common in the development and innovation sectors, where the "hackathon" brand is used to signal modernity and participatory approaches. Mislabelling matters: it creates unrealistic expectations about outputs ("we organised a hackathon, where are

the AI models?"), confuses participants about what they are signing up for, and dilutes the format's credibility.

Common cases of mislabelling include:

- Design thinking workshops that produce personas and journey maps but no functional prototype → these are *créathons* or *design sprints*
- "Innovation days" or "data days" where participants attend presentations and breakout discussions but do not build anything → these are *conferences* or *workshops* – they may produce ideas, brainstorm outputs, or recommendations, but no functional deliverable
- Demo days or pitch competitions where existing projects are showcased and judged → these are *showcases*, not hackathons
- Multi-day bootcamps with tutorials and guided exercises but no competitive element → these are *training programs*
- Créathons or design sprints presented as in-person innovation events → these produce concepts and ideas, not functional prototypes
- Startup weekends focused on business model pitching → these prioritise entrepreneurial framing over technical output

Calling any of these a "hackathon" creates confusion about what the format can deliver. When designing a program, the choice of format should be driven by the intended outcome (see table above), not by the label's perceived prestige.

D. Why Hackathons for Startups? The African Ecosystem Context

Africa's digital ecosystem has grown rapidly over the past decade – according to the International Finance Corporation (IFC, *African Startups That Use Disruptive Tech Do Better*, 2024), the continent hosts one of the fastest-growing startup landscapes globally, with significant concentrations in Nigeria, Kenya, South Africa, Egypt, and increasingly in francophone West Africa. Yet African startups building AI and data-driven products face a set of structural constraints that make competition-based innovation formats particularly relevant.

Data scarcity and fragmentation. Unlike their counterparts in data-rich markets, African startups often operate in environments where relevant training data is scarce, siloed across institutions, or simply not digitised. Agricultural data – crop yields, soil quality, pest incidence, market prices – is frequently collected by public agencies, NGOs, or research institutions but rarely made accessible to private-sector innovators. Hackathons and data challenges create a structured occasion to unlock, prepare, and share datasets that would otherwise remain inaccessible.

Infrastructure gaps. Unreliable electricity, inconsistent internet connectivity, limited access to cloud computing, and weak API ecosystems increase the cost and

complexity of building data products. In-person hackathons provide a temporary but equalising environment — a shared space with guaranteed connectivity, power, and compute resources — while online challenges on platforms like Zindi allow participation from anywhere with a basic internet connection.

A large but dispersed talent pool. Africa produces a growing number of data scientists, software engineers, and AI practitioners, many trained through MOOCs, bootcamps, and university programs. However, this talent is geographically dispersed and often lacks visibility. Online data challenges function as a continent-wide talent surface: they allow individuals from Abidjan, Nairobi, Lagos, or Tunis to compete on equal terms, build a public track record, and connect with potential employers or partners.

Trust deficits in data sharing. Startups operating in emerging markets face heightened challenges around data trust — customers are cautious about sharing personal data, institutional partners demand proof before collaboration, and cross-border data flows raise regulatory concerns. Hackathons, when well-structured, offer a controlled environment for data sharing: time-limited, governed by clear data-sharing agreements, with anonymisation protocols and explicit licensing terms. This reduces the perceived risk for data providers — particularly startups — and creates precedents for future collaboration.

Ecosystem fragmentation. African tech ecosystems are often described as "islands" — vibrant locally but poorly connected regionally. Hackathons serve as ecosystem connectors: they bring together startups, developers, researchers, investors, and institutional partners around a shared challenge, creating relationships that outlast the event itself. In-person events are especially effective at building cross-sector networks in markets where trust is built through face-to-face interaction.

In this context, hackathons and data challenges are not just innovation tools — they are **ecosystem infrastructure**: mechanisms for surfacing data, revealing talent, building trust, and creating the collaborative habits that AI development on the continent requires.

E. The AI4SU Model: Positioning

The AI4SU program operates in two complementary modalities, each drawing on a different format from the typology above:

- **In-person event hackathons** (3 events — Accra, Nairobi): 2–3 day coding hackathons with team-based prototyping, mentorship, and jury evaluation. These events prioritise ecosystem building, face-to-face collaboration between participants and the partner startup, and hands-on technical support.

- **Online data challenges** (5 challenges – hosted on the Zindi platform): multi-week datathon-format competitions open to data scientists continent-wide, with leaderboard ranking and reproducible ML model outputs. These challenges prioritise scale, model quality, and benchmarking.

The combination is deliberate: online challenges generate the best-performing models from a broad talent pool, while in-person events create the relationships, context, and local ecosystem engagement that make those models operationally useful.

Two design principles cut across both modalities:

- **The startup as data provider.** Unlike most hackathons where the organiser supplies the data, AI4SU challenges are built around data shared by the partner startup itself. The startup is not a spectator or sponsor – it co-designs the challenge and is the direct beneficiary of the outputs.
- **Open-source by default.** All winning models, notebooks, and documentation are published on the AI4SU platform, contributing to a growing library of reusable AI assets for the African agritech ecosystem.

Part 3 details the full methodology.

III. THE AI4SU METHODOLOGY

This part describes the AI4SU methodology in three steps: the rationale for focusing on agritech (3.1), the two hackathon modalities and how they are matched to startup needs (3.2), and the full operational cycle from startup identification to model delivery (3.3).

A. Why AgriTech?

The AI4SU program focuses on agricultural technology – with openings to climate, energy, and food systems – for reasons that make the sector uniquely suited to testing data-driven innovation approaches in African contexts.

Data richness and diversity. Agriculture generates a wide range of data types – soil health, weather patterns, satellite imagery, market prices, farmer profiles, transaction histories – each with different collection methods, formats, and access constraints. This diversity makes agriculture a demanding but representative test case for the data challenges that startups face across sectors.

Open data potential. Compared to sectors like fintech or healthtech, where data is overwhelmingly proprietary and regulated, agriculture offers more pathways to open and shared data: public weather stations, satellite imagery (e.g. Sentinel, Landsat), government crop statistics, and research institution datasets. The gap between this

theoretical availability and actual usability is precisely what the program set out to investigate.

Public policy relevance. Food security, rural employment, export competitiveness, and climate adaptation are policy priorities across all eight countries in the program's scope. Data-driven agricultural solutions operate at the intersection of these agendas, giving the program's findings direct relevance to policymakers and development partners.

Variety of use cases with high impact. The eight hackathons conducted under AI4SU span credit scoring for unbanked farmers, crop classification from satellite imagery, market price forecasting, soil nutrient prediction, practice adoption modelling, and produce recommendation systems – demonstrating that data-intensive agriculture is not a single application but a spectrum of solutions with different data requirements and constraints.

AI as a stress test for data ecosystems. AI is the most data-demanding form of digital innovation – and in contexts where data is scarce, fragmented, and poorly governed, it exposes every weakness in the data ecosystem. This is precisely what makes AI a useful lens for the program: not as an end in itself, but as a stress test that reveals where data infrastructure, sharing mechanisms, and governance frameworks need to be strengthened.

B. Two Modalities, One Framework: Matching Format to Startup Data Maturity

The AI4SU program operates through two complementary hackathon modalities. The choice between them is not arbitrary: it depends on the partner startup's data and AI maturity, the nature of the challenge, and the intended impact.

Event hackathons vs. online data challenges

	Event Hackathon	Online Data Challenge
Duration	2–3 days	4–8 weeks
Participants	50–130 (local recruitment)	700–1 100 (continent-wide, open registration)
Platform	Physical venue with compute resources, mentors, and jury	Zindi platform with automated scoring and live leaderboard
Interaction	Face-to-face mentoring, direct collaboration with the startup team	Forum-based discussion, asynchronous iteration
Primary output	Working prototypes + ecosystem relationships	Ranked, reproducible ML models + benchmarks

Best suited for	Ecosystem building, networking, startups needing hands-on support	Scale, model quality, benchmarking, startups with clean datasets
AI4SU	AfriMash (Accra), FarmSawa ×2 (Nairobi)	Tolbi, agriBORA, Farm to Feed, DigiCow, Rhea

Table 3 – Comparison of the two AI4SU hackathon modalities.

The combination is deliberate: online challenges generate the best-performing models from a broad talent pool, while in-person events create the relationships, local context, and hands-on support that make those models operationally useful for the startup.

The live leaderboard: a competition engine

A distinctive feature of the online data challenges is the **live public leaderboard** on the Zindi platform. The leaderboard updates automatically each time a participant submits a model, creating a real-time competitive dynamic that drives iterative improvement throughout the challenge.

The mechanism works as follows:

- Participants can submit up to **10 models per day**, with a maximum of **300 submissions** over the full competition period.
- Each submission is automatically scored against a **public test set** (approximately 30% of the full test data). The resulting public leaderboard is visible to all participants throughout the challenge.
- At the close of the competition, a **private leaderboard** is revealed, scored against the remaining 70% of the test data. This prevents overfitting to the public leaderboard and ensures that the final ranking reflects genuine model performance.

The effect on participant behaviour is significant: top competitors routinely submit **50 or more models** during a single challenge, iterating on feature engineering, model architecture, and hyperparameter tuning in response to leaderboard signals. On the first AI4SU challenge (Tolbi, crop classification), 866 participants generated over **6 400 model submissions** – an average of more than 7 per participant, with the most active contributors submitting dozens.

This iterative dynamic is one of the key advantages of the online format: it transforms the hackathon from a single-shot exercise into a sustained optimisation process, producing models that have been tested and refined over weeks rather than hours.

Matching modality to startup data maturity

The choice of hackathon modality is closely linked to the partner startup's data and AI maturity. AI4SU developed a simple framework to guide this matching:

Startup profile	Data maturity	Typical challenge	Recommended modality
Exploration	Little or no proprietary data; relies on open or third-party datasets	Open-data exploration, proof-of-concept models	Event hackathon (needs hands-on support to frame the challenge)
Valorisation	Data collected through operations but not yet exploited for AI	Predictive modelling, segmentation, scoring on the startup's own data	Online challenge (clean dataset can be prepared for platform) or event (if data needs significant preparation)
Consolidation	Data pipelines in place, existing models to improve	Optimisation, advanced ML, benchmarking against existing baselines	Online challenge (structured data, clear metrics, benefits from scale)

Table 4 – Startup data maturity profiles and recommended hackathon modality.

This framework is not rigid – several AI4SU hackathons combined elements of both modalities. FarmSawa, for example, participated in two event hackathons and could have benefited from an online follow-up challenge to benchmark models at scale. The key insight is that the modality should serve the startup's needs, not the other way around.

In practice, the co-design process (see section 3.3) is where this matching happens: through structured workshops with the startup and the implementation partner, Digital Africa assesses data readiness, identifies the most impactful challenge framing, and recommends the appropriate format.

FOCUS BOX – Structuring Data Sharing Agreements: The AI4SU Two-Step Approach

Data sharing is the single most sensitive aspect of hackathon organisation with startups. Startups are asked to share proprietary data – customer records,

transaction histories, agricultural datasets – with external participants, in a competitive format, often across borders. Without a robust legal framework, this creates unacceptable risks for the startup: loss of competitive advantage, regulatory exposure, or reputational damage.

At the same time, overly rigid legal processes can kill hackathon momentum. If signing a data sharing agreement takes months of legal review, the startup disengages and the challenge window closes.

AI4SU developed a two-step legal framework, with the support of Cage Patrick & Company, designed to protect the startup without slowing down the innovation process.

Step 1 – Feasibility-stage Data Sharing Agreement (DSA). Signed early in the co-design phase between Digital Africa, the startup, and Zindi. This agreement enables controlled, internal access to the startup's data for the sole purpose of assessing technical feasibility and preparing the dataset for competition. At this stage, data is not shared publicly – it remains within the preparation team (Digital Africa + Zindi). The DSA covers:

- Scope of data access (what data, for what purpose, for how long)
- Anonymisation and pseudonymisation requirements
- Confidentiality obligations for all parties
- Data deletion protocols after the feasibility assessment

This step allows the team to validate whether the startup's data is suitable for a hackathon challenge before committing to a full competition – reducing the risk of launching a challenge that fails due to data quality issues.

Step 2 – Cooperation agreement (competition framing). Signed once feasibility is confirmed and the dataset has been prepared. This agreement frames the conditions under which the prepared dataset is released to hackathon participants on the Zindi platform. It covers:

- Licensing terms for the dataset (typically under Creative Commons or equivalent open-data terms)
- Intellectual property: participants waive IP on submitted models, which are published in open-source
- Prize structure and distribution modalities
- Visibility and branding obligations
- Post-competition data handling (retention, deletion, archiving)

Why two steps? The staged approach addresses the core tension between startup protection and innovation speed:

- Startups are more comfortable sharing a limited dataset at the feasibility stage, when the scope is narrow and the audience is restricted. This lowers the barrier to entry.
- If feasibility is not confirmed (e.g. the data is insufficient or too sensitive), the process stops without the startup having exposed its data publicly.
- Once feasibility is confirmed, the cooperation agreement can be negotiated on the basis of a dataset that has already been reviewed, cleaned, and anonymised – making the legal conversation faster and more concrete.

For event hackathons delivered through implementation partners (e.g. Impact Hub Nairobi, Blossom Academy), a simplified DSA arrangement was used, adapted to the shorter timeline and smaller participant pool.

The DSA templates were iterated throughout the program based on startup feedback and operational learning from each hackathon cycle, with the objective of reducing friction while maintaining adequate protection.

C. The AI4SU Cycle: From Co-Design to Diffusion

The AI4SU methodology follows a three-phase operational cycle – **co-design** → **hackathon** → **diffusion** – that structures the entire engagement with each partner startup from first contact to final delivery. Understanding this cycle is essential because each phase depends on the outputs of the previous one: shortcuts in co-design produce poorly framed challenges; poorly framed challenges produce unusable models; unusable models mean no diffusion.

Startup identification

Before co-design begins, the program must find the right startups. AI4SU used multiple sourcing channels:

- **Portfolio mining:** Digital Africa's and Proparco's existing portfolios of agritech investments and supported startups
- **LinkedIn sourcing:** targeted outreach to founders and CTOs of data-intensive agritech startups across the program's focus countries
- **Events and conferences:** agritech and AI events – including field missions in Dakar, Nairobi, and Marrakech where the team met startups face-to-face. These field visits also served as a validation channel: meeting startup teams in person helped verify data capacity claims from surveys and build the trust needed for subsequent data-sharing discussions
- **Incubator and accelerator networks:** partner incubators and hubs across the eight countries

No single channel was sufficient. LinkedIn surfaced startups outside any institutional portfolio; field visits identified locally active startups with no online visibility; portfolio mining revealed startups already vetted but not yet engaged on data-sharing topics. A critical function of the sourcing phase is **early assessment of each startup's data capacity** – determining whether the startup holds enough shareable data to sustain a viable competition before investing significant time on either side. There is no value in engaging a startup through a months-long co-design process only to discover that its data assets are insufficient for a hackathon challenge. Screening for data readiness upfront protects the startup's time as much as the program's. Broad sourcing also pays dividends at the other end of the cycle: the full pool of startups identified during the program – including those not ultimately selected for a hackathon – forms a **natural diffusion network** for the results. Sharing open-source models, documentation, and platform links with all sourced startups extends the program's impact well beyond the direct partners and reinforces the ecosystem logic of the initiative.

Three co-design methods tested

AI4SU did not apply a single co-design methodology. Three distinct approaches were tested, each with different characteristics in terms of co-design format, ecosystem engagement, and timeline. The methods are named here by their defining methodology, not by the implementation partner – though each was piloted with a specific partner (noted in parentheses).

Method 1 – Deep-dive pathway (piloted with Zindi, online data challenges)

The most structured of the three approaches, used for the five online challenges. Digital Africa and Zindi worked one-to-one with each selected startup, moving from a broad data needs survey to a fully prepared competition dataset through a series of focused interactions:

The process follows six steps:

1. **Data needs survey** sent to pre-screened startups, capturing data assets, AI challenges, and willingness to share data
2. **Joint feasibility analysis:** Digital Africa analysed survey responses with Zindi, assessing both the technical feasibility of the challenge (data quality, volume, structure) and its relevance to the program and the broader ecosystem
3. **Co-design workshops** (1 to 4 per startup, depending on data complexity and needs): co-facilitated by Digital Africa and Zindi with each selected startup. Zindi arrived with concrete challenge proposals derived from the survey responses – specific problem statements, target variables, evaluation metrics. The workshops also addressed legal requirements and introduced the Data Sharing Agreement framework (see Focus Box above)

4. **DSA signature → data preparation:** once the feasibility-stage DSA was signed, the preparation team accessed the startup's data, confirmed feasibility definitively with Zindi, and prepared the competition dataset (cleaning, anonymisation, train/test split). This phase concluded with a **final challenge brief validated by the startup** before launch
5. **Challenge launch:** 4–8 week online competition on the Zindi platform
6. **Model audit → restitution:** Zindi audited the winning models for reproducibility and quality; the top solutions across the three prize categories and their documentation were then presented to the startup

This pathway is defined by its one-to-one co-design channel with a single startup, a high technical feasibility bar (the dataset must sustain a multi-week competition on an automated platform with hundreds of participants), and the need to build a robust trust framework for data sharing through the DSA process (see Focus Box above).

Method 2 – Ecosystem-driven pathway (piloted with Impact Hub, event hackathons, Nairobi)

A broader approach, used for the two Nairobi events. Impact Hub mobilised its local network to bring together not just startups but also data producers, institutional stakeholders, researchers, and local tech communities – casting a wider net than any single startup's perspective:

1. **Multi-stakeholder co-design workshop:** a large upstream workshop bringing together agritech startups, data producers, institutional stakeholders, and ecosystem actors to identify thematic challenge areas in agritech – casting a wider net than any single startup's perspective
2. **Webinar cycle:** a series of 8 focused webinars, each dedicated to refining a specific challenge theme with relevant startups and domain experts. These webinars served a dual purpose: structuring the challenge content and **building anticipation** for the hackathon among the local community. Not all themes were carried forward to the competition – only **3 challenges** were ultimately selected for the hackathon, based on data readiness and ecosystem relevance
3. **Event hackathon:** 2–3 day in-person hackathon with the selected challenges

This method required significant convening capacity and a strong local network, but it surfaced challenge areas that no single startup would have identified alone – and created a mobilised community before the hackathon even began. Its defining feature is the **collective, multi-stakeholder co-design**: multiple startups and ecosystem actors participate in the upstream phase, even though ultimately only one startup's data is used for each hackathon challenge.

Method 3 – Direct outreach pathway (piloted with Blossom Academy, event hackathon, Accra)

The most direct of the three approaches, used for the Accra event. Blossom Academy leveraged its existing relationships in the Ghanaian and regional tech ecosystem to identify and engage startups rapidly:

- I. **Targeted outreach:** the implementation partner contacted startups directly by email and phone, leveraging existing relationships in the local and regional tech ecosystem – including cross-border startups that formal sourcing channels might not have reached
- II. **One-on-one co-construction:** challenge definition through direct brainstorming sessions between the partner's team and the startup – closer to a consulting engagement than a structured workshop process
- III. **Event hackathon:** 2–3 day in-person hackathon

This method can move from first contact to hackathon in weeks rather than months. Its defining features are speed (built entirely on **pre-existing trust relationships**), cross-border reach (the partner's network enabled targeting startups outside the host country), and minimal infrastructure requirements – but it does not scale beyond one or two startups per cycle.

Cross-method lessons

- The **deep-dive pathway** is the most replicable: any program with access to a competition platform and basic data science capacity can apply it, and it can operate fully online without a local anchor. Its weakness is timeline: the full cycle from survey to restitution takes 4–6 months.
- The **ecosystem-driven pathway** is the most inclusive: the upstream multi-stakeholder workshop surfaces challenges and data sources that targeted sourcing would miss. Its weakness is dependence on a strong local convener.
- The **direct outreach pathway** is the fastest: when the trust relationship exists, co-design can happen in weeks rather than months. Its weakness is that it does not scale – it works for one or two startups, not for a portfolio.
- All three methods converge on the same principle: **the challenge must be derived from real data and real operational needs**, not from theoretical problem statements. The path to that data differs; the destination does not.

Evaluation and prize design

How solutions are evaluated and rewarded differs fundamentally between the two modalities – and this difference matters for how startups receive and use the outputs.

Online data challenges – automated performance scoring:

- Solutions are ranked through the Zindi **leaderboard** (see section 3.2): automated scoring against a public test set during the competition, then a private leaderboard revealed at close to determine final rankings

- The evaluation is purely quantitative: model performance on the defined metric (RMSE, F1 score, accuracy, etc.)
- Prize categories: best overall model, best model by a woman or team including a woman, best model by a local participant (i.e. from the challenge's country focus)

Event hackathons – jury evaluation with scoring grid:

- Solutions are evaluated by a jury panel using a structured scoring grid that assesses multiple dimensions: technical quality, innovation of the approach, business relevance to the startup, presentation quality, and documentation
- The evaluation is qualitative and multi-criteria – a technically strong model that ignores the startup's operational constraints may score lower than a simpler but more deployable solution
- Prize categories: best overall solution, best model by a woman or team including a woman, most innovative approach (replacing the "best local" category, which is less relevant when participants are primarily local)

This difference in evaluation reflects the fundamental distinction between the two modalities: online challenges optimise for **model performance at scale** (hundreds of participants, automated benchmarking), while event hackathons optimise for **solution relevance and ecosystem building** (face-to-face interaction, jury deliberation, multi-criteria assessment).

Diffusion

The final phase transforms hackathon outputs into reusable ecosystem assets:

- **Model audit:** top solutions are reviewed for reproducibility, code quality, and documentation completeness (by Zindi for online challenges, by the organising team for events)
- **Delivery workshops:** structured handover sessions with the startup (see Best Practice #8, Part 4)
- **Open-source publication:** all winning models, notebooks, and documentation are published on the [AI4SU platform](#) with standardised metadata and open licensing
- **Winner-startup connections:** the program facilitates direct contact between top participants and the partner startup for potential follow-up collaboration or hiring

Without structured delivery and publication, hackathon outputs remain private files on participants' laptops – technically open but practically inaccessible. The diffusion phase is where the program's open-source commitment is operationalised.

IV. BEST PRACTICES

The following best practices are drawn from the operational experience of the AI4SU program. They are organised along the three phases of a hackathon cycle – design & preparation, execution, and diffusion & impact – and are intended as actionable guidance for program managers seeking to organise hackathons with and for startups.

A. Phase 1 – Design & Preparation

Best Practice #1 – Identify the Right Startups: Survey, Screen, Don't Over-Solicit

Startups in emerging ecosystems are small teams, often solicited by multiple programs simultaneously. "Program fatigue" is real: founders receive dozens of partnership proposals, grant applications, and collaboration requests. Engaging the wrong startup – or engaging the right one too aggressively – wastes time on both sides.

The first step is therefore not outreach but **screening**. AI4SU developed a structured pre-screening survey focused on startups' data assets and AI needs. The survey served a dual purpose:

- **Filtering:** it identified startups with genuine data challenges suited to the hackathon format, avoiding the cost of engaging startups whose data was insufficient, too sensitive, or not relevant. This also protected startups from investing time in a process that would not lead to a viable challenge.
- **Preparation:** survey responses allowed the Digital Africa and Zindi teams to arrive at the first co-design workshop with preliminary insights and challenge proposals – not with a blank page. This shifted the dynamic from brainstorming to co-design, which startups found more respectful of their time.

The two-step DSA framework (see Part 3, Focus Box) reinforced this approach: by securing a feasibility-stage agreement early, the team could explore the startup's data during the preparation phase, assess whether a challenge was viable, and avoid the scenario of investing months of co-design only to discover that legal constraints made the hackathon impossible.

Key takeaways

- **Screen before you engage.** A structured survey saves time for both the program and the startup. Not every startup is suited to a hackathon – identifying the right candidates early is a best practice, not a barrier.
- **Respect startup time constraints.** Avoid long email threads and multi-step administrative processes. Prefer direct conversations – a 30-minute call achieves more than ten emails.

- **Include both founders and technical teams from the start.** Founders provide strategic context and decision-making authority; technical teams (CTOs, data engineers) provide the operational reality of data availability. If only one side is present, critical information is missed or delayed.
- **Expect uneven communication.** Startup founders are juggling multiple priorities. Responsiveness is a signal of engagement – but inconsistent replies are normal, not a sign of disinterest. Prioritise startups that respond with both speed and rigour, but build buffer time into the co-design timeline.
- **Start co-design at least 8–12 weeks before the hackathon,** but structure the process into a small number of focused touchpoints rather than a continuous stream of meetings.
- **Treat data capacity as a GO/NO-GO filter.** A startup with a compelling business case but insufficient shareable data will not produce a viable hackathon challenge. Early assessment of data readiness – volume, quality, shareability – is the single most important screening criterion (see section 3.3, Startup identification).

Best Practice #2 – Co-Design the Challenge from Real Data, Not Theory

The single most common failure mode in hackathon design is starting from a theoretical challenge rather than from the startup's actual data and operational needs. A well-framed academic problem – "predict crop yield from satellite imagery" – sounds compelling but may be impossible to implement if the startup does not have the right data, the right data format, or the operational capacity to integrate the resulting model.

AI4SU's co-design process reversed this logic: **start from the startup's data and pain points, then derive the challenge.** Section 3.3 describes the three co-design methods tested in detail; the principles below apply across all three.

In practice, this meant:

- **Real dataset inspection, not theoretical assessment.** During the feasibility stage (enabled by the Step 1 DSA), the Zindi team conducted a hands-on review of the startup's actual data – structure, quality, volume, missing values, potential biases. This was not a questionnaire about data assets; it was a technical audit of the data itself.
- **Challenge framing workshops.** Based on the data audit, the Digital Africa and Zindi teams proposed challenge framings to the startup – specific problem statements, target variables, evaluation metrics – that were grounded in what the data could actually support. The workshops were structured as co-design sessions, not brainstorm: the team arrived with concrete proposals, and the startup refined them based on operational context.

- **Modality matching.** The data maturity framework (see Table 4, Part 3) guided the choice between event hackathon and online data challenge. Startups at the "Exploration" stage – with limited proprietary data – benefited more from the hands-on support of an in-person event, while startups at the "Valorisation" or "Consolidation" stages could prepare clean datasets for the Zindi platform and leverage the scale of an online challenge.

This approach had a critical downstream benefit: challenges built from real data produce models that the startup can actually use. When the problem statement matches the startup's operational reality and the training data reflects its actual data environment, the winning models are not academic exercises – they are candidate solutions for deployment.

Key takeaways

- Never design a challenge without inspecting the actual data. Theoretical challenges lead to theoretical outputs. A hands-on data audit during feasibility is non-negotiable.
- Co-design, don't brainstorm. Arrive at workshops with concrete proposals based on the data audit. Startups respond better to structured options than to open-ended ideation.
- Match the challenge to the startup's capacity to reuse the models. A sophisticated model that the startup cannot integrate is a wasted output. Frame challenges with deployment in mind.
- Adapt the competition modality to the data reality, not to program convenience. The format should serve the startup's needs (see Part 3, section 3.2).
- Use the data maturity framework (Table 4) as a decision tool. It provides a shared language for matching startups to the right hackathon format – and helps avoid the common mistake of forcing all startups into the same modality.
- Expect 1 to 4 co-design workshops per startup. The depth of engagement depends on data complexity and the startup's readiness – the process is not one-size-fits-all.

Best Practice #3 – Secure Data Sharing Early with a Staged Legal Framework

Data sharing is the critical enabler – and the most common blocker – of hackathons with startups. Without a clear legal framework, startups will not share the data that makes the hackathon possible. But if the legal process is too rigid or too slow, the co-design momentum is lost and the startup disengages.

AI4SU's key insight was to **stage the legal process** rather than front-load it. The two-step DSA framework (detailed in Part 3, Focus Box) separates feasibility-stage data access from competition-stage data release:

- **Step 1 – Feasibility DSA:** signed early in the co-design phase, enabling the preparation team to access the startup's data internally for assessment purposes only. This allows the team to verify data quality and challenge viability before committing to a full competition – without exposing the startup's data publicly.
- **Step 2 – Cooperation agreement:** signed once feasibility is confirmed, framing the conditions for public dataset release, IP terms, prize structure, and post-competition data handling.

The staged approach directly addresses the most common failure scenario: investing months of co-design only to discover that legal or data constraints make the hackathon impossible.

For event hackathons delivered through local implementation partners, a simplified DSA arrangement was used, adapted to the shorter timeline and smaller participant pool.

Key takeaways

- Sign the feasibility-stage DSA before intensive co-design begins. Early legal clarity protects both the program and the startup from wasted effort.
- Adapt the legal framework to the modality. Online challenges require a full two-step process (broader data exposure, longer timeline); event hackathons can use a simplified arrangement.
- Budget 4–6 weeks for legal processes. Data-sharing agreements involve founders, legal teams, and sometimes board-level decisions. Do not assume this can be compressed into a single meeting.
- Iterate the templates. AI4SU refined its DSA templates after each hackathon cycle based on startup feedback – the first version is never the final one.

B. Phase 2 – Execution

Best Practice #4 – Prepare the Data as a Product

The quality of hackathon outputs is directly determined by the quality of the data provided to participants. Poorly documented, messy, or incomprehensible datasets lead to participants spending most of their time on data wrangling rather than model

development – and produce solutions that are difficult for the startup to evaluate or reuse.

AI4SU treated dataset preparation as a **product development exercise**, not an administrative task. For each technical challenge, try to produce the following documents during the preparation phase:

- A **structured data dictionary**: every variable documented with its name, type, unit, description, expected range, and proportion of missing values. This is the single most impactful preparation artefact – participants who understand the data build better models.
- A **clear problem statement with evaluation metrics**: the target variable, the scoring metric (e.g. RMSE, F1 score, accuracy), and the business context that explains *why* this metric matters for the startup.
- A **baseline model**: a simple, documented reference model (e.g. logistic regression, random forest with default parameters) that participants could use as a starting point. Baselines set expectations and reduce the barrier to entry for less experienced participants.
- A **pre-populated FAQ**: anticipated questions about the data ("What does column X represent?", "Why are there missing values in Y?", "What time period does the data cover?") answered proactively. This reduced repetitive forum queries and accelerated participant onboarding.
- **Online discussion forum**: Treat the forum as part of the deliverable. Define who answers what (platform support vs. startup domain experts), set a response routine (e.g., daily check-ins), and pin one "FAQ + updates" thread to reduce duplicate questions.

The preparation requirements also differ by modality. For **online challenges**, the dataset must be fully self-contained: every ambiguity not addressed in the documentation will generate forum questions at scale, with no real-time support to compensate. For **event hackathons**, the presence of on-site mentors and the startup's own team provides a live clarification channel – the dataset can be less polished, as long as domain experts are available to fill the gaps.

Key takeaways

- Document everything. Dataset description, variable definitions, objectives, scoring system, known limitations. The more precise the documentation, the more the solutions will be adjusted to the startup's actual needs.
- Provide a baseline model. It lowers the barrier to entry and gives participants a concrete starting point.

- Keep the startup available. Participants need access to domain expertise – schedule regular Q&A sessions or forum check-ins, particularly during the first week of an online challenge.
- Invest in the FAQ. Every hour spent anticipating participant questions saves dozens of hours of redundant forum exchanges.

Best Practice #5 – Structure Mentorship and Technical Support

Participants need two kinds of support to produce usable outputs: **domain context** (understanding the startup's business, the agricultural context, the end-users) and **technical guidance** (data science methodology, platform mechanics, submission formatting). Without structured access to both, even talented participants produce models that are technically sound but operationally irrelevant.

The support model differs by modality:

For event hackathons:

- **Startup team on-site.** The partner startup's technical and business teams can be physically present throughout the event, available for direct questions and feedback. This real-time interaction is one of the key advantages of the in-person format – participants could ask "what does this column mean?" and get an immediate, contextualised answer.
- **Mentor rotations.** Domain experts and senior data scientists rotated between teams on a structured schedule, providing targeted guidance without monopolising any single team's time.
- **Peer engagement.** Event hackathons naturally fostered collaboration and knowledge-sharing between participant teams – an emergent dynamic that the organisers encouraged through shared workspaces and informal networking moments.

For online challenges:

- **Zindi team support.** The Zindi team monitored the forum, responded to technical questions about the platform and submission format, and escalated domain-specific questions to the startup.
- **Structured Q&A sessions.** For longer challenges, scheduled "Ask Me Anything" sessions with the startup's data team provided a concentrated window for participants to clarify data-related questions.
- **Community-driven support.** Experienced Zindi competitors often answered questions from less experienced participants on the forum, creating a self-reinforcing support dynamic.

The **webinar cycle** used in the ecosystem-driven pathway (see section 3.3) offers an additional support model: a series of focused online sessions before the hackathon that simultaneously refine the challenge content and build a community of engaged potential participants. This pre-hackathon engagement means participants arrive better prepared – reducing the support burden during the event itself.

Key takeaways

- For events, make the startup team physically present and accessible – not just for the opening keynote, but throughout the hackathon.
- For online challenges, schedule regular touchpoints between participants and the startup's domain experts if possible.
- Leverage peer support. Experienced participants are a support resource – create the conditions for them to help others (forum, shared channels, team mixing).

Best Practice #6 – Attract the Right Participants: Recruit by Modality

The quality of hackathon outputs depends directly on the quality – and the fit – of the participants recruited. Data scientists form a distinct community with specific habits, motivations, and availability constraints. Recruiting them requires understanding who they are, where they are, and what drives them – and the approach differs fundamentally between the two modalities.

Online data challenges – platform communities first:

Data scientists who compete on platforms like Zindi, Kaggle, or DrivenData follow leaderboards, build public profiles, and select challenges based on technical interest and prize value. Standard program communication channels (newsletters, institutional websites) do not reach them. The most effective recruitment channel is **platform-native promotion**: challenges promoted through Zindi's existing user base, mailing lists, and social media channels. Word-of-mouth among competition regulars – who recruit teammates and share challenges in their networks – amplifies reach organically. Because your challenge competes with many other competitions, it is important to calibrate prize amounts to the technical difficulty and time required, and avoid both over- and under-remunerating participants. Online challenges attract a predominantly **professional and semi-professional** data science audience, often competing alongside full-time employment. The asynchronous format accommodates this – participants work on their own schedule over several weeks.

Event hackathons – local ecosystems and the working-day constraint:

In-person events require physical availability for 2–3 consecutive days – a significant constraint that shapes who can participate. AI4SU observed that many participants at event hackathons – particularly at Impact Hub Nairobi – took **personal leave from their jobs** to attend. This highlights both the perceived value of the events and a structural barrier: participation depends on employment flexibility and personal sacrifice. The participant mix at events is broader than online: alongside experienced data scientists, events attract **students, junior developers, and domain experts** (agronomists, business analysts) who would not compete in an online data challenge but bring valuable contextual knowledge. Recruitment for events relies heavily on **local implementation partners** (hubs, universities, tech communities) who have the trust and networks to mobilise participants. The webinar cycle used in the ecosystem-driven pathway (see section 3.3) also served as a recruitment and anticipation-building tool.

Cash prizes as a participation driver:

Across both modalities, **cash prizes** are a significant motivator – particularly in African markets where data science talent is growing but economic opportunities are uneven. Prize amounts must be calibrated to be meaningful in the local context while remaining within program budgets. Beyond cash, non-monetary incentives matter: visibility on the platform leaderboard (online), certificates, networking opportunities with the startup and program partners (events), and the possibility of follow-up collaboration or employment.

Key takeaways

- Recruit through the channels where data scientists already are – competition platforms and tech communities, not institutional newsletters.
- Account for the working-day constraint when planning event hackathons. Weekend or holiday scheduling can widen the participant pool; weekday events require participants to take personal leave, which signals engagement but limits reach.
- Leverage local partners for event recruitment – they have the trust and networks that program-level outreach cannot replicate.
- Use cash prizes strategically. They attract participants, but the amount and structure (multiple categories vs. winner-take-all) shape who competes and how.

C. Phase 3 – Diffusion & Impact

Best Practice #7 – Require Documented Solutions, Not Just Code

A winning model is only as valuable as its documentation. Code that runs but cannot be understood, reproduced, or adapted by the startup's own developers is a dead-end artefact. The goal is not to produce the best-performing model on a leaderboard – it is to produce solutions that the startup and the broader ecosystem can learn from, adapt, and build upon.

AI4SU integrated solution documentation as a formal requirement, not a bonus:

- **Structured solution write-ups.** Top participants were required to submit a written description of their approach. This documentation can cover: data preprocessing steps, feature engineering choices, model architecture, hyperparameter tuning, and – critically – **what they tried that did not work**. This last element is particularly valuable: documenting failed approaches prevents other developers from repeating the same dead ends and provides the startup with a fuller picture of the solution landscape.
- **Reproducible notebooks.** Winning solutions were published as fully executable notebooks, with clear code comments and step-by-step explanations. The standard was not "does it run?" but "can another data scientist understand and modify it?"
- **Multiple solutions, not just the winner.** AI4SU published not only the top-ranked model but the top 3 solutions for each challenge. This was a deliberate design choice: startups benefit from seeing **different approaches to the same problem** – different feature engineering strategies, different model architectures, different trade-offs between accuracy and interpretability. A startup may ultimately build its production model by combining elements from several competition solutions.

Documenting what did not work is particularly important ; failed experiments are valuable information for the ecosystem – they help avoid the **survivorship bias** that plagues most hackathon reporting.

Key takeaways

- Make solution documentation a deliverable, not an afterthought. Include it in the challenge rules and evaluation criteria.
- Document failures as well as successes. What did not work is as informative as what did – especially in data-scarce environments.
- Publish multiple solutions. The startup needs to see different ways to approach the problem, not just the single best-performing model.

- Consider a "Best Documented Solution" prize to incentivise quality write-ups alongside model performance.

Best Practice #8 – Bridge the “Last Mile” from Model to Operations

The gap between a winning model on a leaderboard and a deployed model in production is one of the most underestimated challenges in hackathon-based innovation. Without a structured post-hackathon process, even excellent models remain academic artefacts – technically impressive but operationally useless.

AI4SU addressed this through several mechanisms:

- **Delivery workshops.** After each online hackathon, Digital Africa organised a structured handover session with the startup, walking through the top solutions, explaining the approaches, and identifying which elements could be integrated into the startup's existing technical stack.
- **Winner-startup connections.** The program facilitated direct contact between winning participants and the partner startup. In several cases, this created opportunities for follow-up collaboration – participants with deep knowledge of the startup's data and challenge became valuable technical contacts.
- **Prize design as a strategic tool.** AI4SU's prize structure was intentionally diversified – and adapted to each modality (see section 3.3). For online challenges: best overall model, best model by a woman or team including a woman, and best model by a local participant. For event hackathons: best overall solution, best by a woman or team including a woman, and most innovative approach – reflecting the jury-based, multi-criteria evaluation where local participation is already the default. In both cases, the logic is the same: rewarding only the single best-performing model narrows the startup's options; rewarding diverse approaches enriches them.

The underlying insight is that **the talent is as valuable as the code**. Top hackathon participants are skilled data scientists who have spent weeks understanding the startup's data and problem. Losing contact with them after the competition is a missed opportunity – for the startup (which could hire or contract them) and for the ecosystem (which benefits from retaining and developing this talent).

Key takeaways

- Plan post-hackathon delivery from the start. A structured handover workshop should be part of the hackathon timeline, not an afterthought.

- Connect winners with the startup. The participants who built the best models are the ones best positioned to help integrate them.
- Diversify prize categories. Don't reward only the top-performing model – reward diversity of approaches, documentation quality, and inclusivity.
- Think beyond the model. The talent, the community, and the documented approaches have long-term value that exceeds the immediate competition output.

Best Practice #9 – Clarify IP and Cultivate Returning Participants

Intellectual property (IP) clarity is a non-negotiable prerequisite for any hackathon producing open-source outputs. Open-source publication of models is a core principle of the AI4SU program – but participants need to understand and accept this before they start working. IP terms must be explicit in the challenge rules, the cooperation agreement, and the platform terms of service. Ambiguity about IP creates legal risk, discourages experienced participants who have been burned by unclear terms in other competitions, and undermines the program's open-source credibility.

An underexploited dynamic is participant retention across challenges. Participants who competed in the Tolbi challenge and then returned for agriBORA or DigiCow brought cross-domain techniques and accumulated platform experience. Zindi profiles track this history, creating a visible track record that benefits both the participant and the ecosystem. Programs that run multiple challenges should actively cultivate this returning participant base – it is a community asset that compounds over time.

Key takeaways

- Clarify IP terms upfront. Open-source commitments must be explicit, visible, and accepted before the challenge begins.
- Cultivate returning participants. Cross-challenge retention builds a community of practice that improves model quality over time.

D. Perspectives: Beyond the AI4SU Model

The AI4SU model – startup as data provider, co-designed challenge, open-source outputs – is one configuration among several. Several alternative approaches and variants merit consideration for future programs:

Hackathons on third-party data with startup application. Instead of starting from a single startup's data, a program could partner with research institutions, public agencies, or open-data providers to prepare a dataset, then issue an open call for startups to propose use cases based on that data. This inverts the AI4SU model: the data comes first, and the startup brings the business application. The advantage is that it removes the data-sharing barrier entirely – the dataset is already available – and opens the challenge to startups that may not have proprietary data but have strong domain expertise and operational capacity to deploy models.

Hackathons as talent identification and recruitment. Several AI4SU partner startups noted that the hackathon process gave them unprecedented visibility into the skills of individual data scientists – not through CVs or interviews, but through **demonstrated performance on their own data**. This positions hackathons as a powerful recruitment tool: startups can observe candidates solving a real problem with real data under competitive pressure, assess code quality and documentation practices, and identify top talent for potential hiring. For startups in African markets where the data science talent pool is growing but hard to assess through traditional channels, this is a significant secondary benefit of the hackathon format.

Hackathons as monitoring and evaluation infrastructure. For development programs and DFIs, hackathons offer an unusual advantage: they produce **quantifiable, comparable outputs**. Model accuracy scores, number of submissions, participant demographics (country, gender, experience level), geographic reach – these are objective indicators that can feed directly into program M&E frameworks. Unlike many innovation activities where impact is measured through self-reported surveys or narrative case studies, hackathon outputs are measurable by design. This makes the format particularly attractive for donors seeking evidence-based reporting on innovation investments.

Call-for-applications model. The AI4SU use case identification work (see companion Use Cases report) suggests another variant: rather than co-designing a challenge with a pre-selected startup, a program could define challenge themes based on sector-wide use case analysis and issue an **open call for startups to apply** to benefit from a hackathon on their data. This inverts the sourcing logic – startups self-select based on the published themes. The approach is less co-design-intensive but could dramatically expand the pool of potential partners, particularly for programs operating across many countries or sectors.

Multi-startup hackathons. A largely unexplored variant is organising a single hackathon around **data from multiple startups simultaneously** – either as competing challenges or as complementary datasets. This would increase the hackathon's ecosystem impact but introduces significant design complexity: challenge framing must avoid creating competitive tensions between participating startups, and data-sharing agreements become multi-party arrangements. AI4SU did not pursue

this model, but it remains an interesting frontier for programs with strong challenge design capacity.

V. CONCLUSION

The AI4SU program was a practical approach to applied data governance: using hackathons not as standalone events but as structured instruments for unlocking data, building AI capacity, and connecting startups with the talent they need. Over eight hackathons, more than 4 700 participants, and 24 open-source models, the program generated a body of operational experience that extends well beyond the agritech sector. Five key learnings emerge.

A. Key Learnings

1. *Startups are the right actors for hackathon-based innovation – but they are fragile partners.*

Startups bring what no other actor can: real operational data, genuine business problems, and the agility to integrate solutions quickly. They ground hackathon challenges in the real economy – a credit scoring model built on FarmSawa's actual farmer data is fundamentally different from one built on a synthetic dataset. But startups are also small teams with limited bandwidth, competing priorities, and no obligation to participate in external programs. They can – and do – build AI products without hackathons. A program's value proposition must be clear and tangible: access to talent, open-source models, structured data preparation support, R&D boost. The moment a startup feels the process is extractive rather than generative, it disengages.

2. *Data quality determines everything downstream.*

This is the single most consistent finding across all eight hackathons. When the challenge is built from real, well-prepared data – inspected during feasibility, cleaned and documented as a product – the resulting models are deployable. When shortcuts are taken in data preparation, the outputs are academic exercises. The entire upstream methodology (co-design workshops, data maturity framework, feasibility-stage DSA) exists to ensure that by the time a challenge launches, the data is ready. Every hour invested in data preparation saves days of wasted participant effort and produces models the startup can actually use.

3. *Data sharing requires staged trust, not just legal frameworks.*

The two-step DSA approach is perhaps the program's most transferable contribution. Data sharing between startups and external participants is not a legal checkbox – it is a trust-building process. The staged framework (feasibility DSA first, cooperation agreement second) acknowledges this: it gives startups a low-risk entry point, validates feasibility before public exposure, and builds

confidence incrementally. Programs seeking to replicate this model should invest as much in the trust architecture as in the technical infrastructure.

4. *A hackathon produces working outputs – if that is not your goal, choose another format.*

The proliferation of events labelled “hackathons” in the development sector has diluted the term to the point of confusion. A hackathon is defined by four criteria: time pressure, a defined challenge, a working output, and competitive evaluation. If the objective is ideation, use a design sprint. If the objective is networking, organise a conference. If the objective is training, run a bootcamp. Mislabelling creates unrealistic expectations (“where are the AI models?”) and wastes resources. The AI4SU experience confirms that when a genuine hackathon is well-designed, it delivers measurable, reusable outputs – but only because the format was matched to the objective.

5. *Hackathons are ecosystem infrastructure, not just events.*

The value of a hackathon extends far beyond the models produced during the competition. AI4SU hackathons surfaced data science talent across the continent (BP#6), created direct connections between winners and startups for potential hiring or collaboration (BP#8), built open-source assets accessible to the entire ecosystem, and generated a returning community of participants who improved with each cycle (BP#9). For donors and DFIs, this means that the return on investment of a well-structured hackathon program is systemic, not transactional: it strengthens the ecosystem's capacity to produce and use AI, one challenge at a time.

B. Conditions for Replicability

The AI4SU methodology is not sector-specific. The data maturity framework, the co-design cycle, the DSA architecture, and the dual-modality model can be adapted to healthtech, fintech, climate, logistics, or any domain where startups hold underexploited data assets. Four conditions are necessary for successful replication:

Access to a competition platform. Online data challenges require a technical platform with automated scoring, leaderboard management, and dataset hosting. Platforms like Zindi, Kaggle, or DrivenData provide this infrastructure out of the box. For programs without platform access, the event hackathon modality remains viable with lighter infrastructure – but at the cost of scale and benchmarking capacity.

Data science capacity in the preparation team. At least one data science or machine learning profile is needed within the program team to conduct the feasibility-stage data audit, prepare the competition dataset (cleaning, anonymisation, train/test split), build baseline models, and assess the quality of submitted solutions. Without this capacity, the program depends entirely on external partners for technical judgment.

A legal framework for data sharing. The DSA templates developed under AI4SU – with the support of Cage Patrick & Company – are adaptable to other jurisdictions and sectors. The key principle is staging: separate feasibility-stage access from competition-stage release. Programs operating in new regulatory environments should budget for local legal review and expect to iterate the templates over multiple cycles.

Local implementation partners for event hackathons. In-person events require a convening partner with established relationships in the local tech ecosystem – a hub, university, or tech community organisation that can recruit participants, secure venues, and provide on-the-ground coordination. The ecosystem-driven and direct outreach pathways tested by AI4SU both depend on this local anchoring. Online challenges, by contrast, can be managed centrally.

The methodology is also **sector-transferable**. Agriculture was chosen for AI4SU because of its data diversity, open-data potential, and policy relevance – but the same co-design process, maturity framework, and hackathon architecture apply wherever startups hold operational data that could benefit from external AI development capacity.

C. Open Questions for Future Iterations

Three questions remain open for programs seeking to build on the AI4SU experience:

How to scale beyond one-startup-at-a-time?

The AI4SU model is resource-intensive: each hackathon involves months of co-design with a single startup. Scaling to dozens of startups per cycle would require a different sourcing and co-design architecture. Two paths merit exploration. First, the **call-for-applications model**: defining challenge themes based on sector-wide use case analysis, then issuing an open call for startups to apply. The sourcing survey could be adapted to ask startups whether they would be interested in collective modalities, and a collective workshop could be organised to present the call and co-design challenge themes with multiple startups simultaneously. Second, adapting the **ecosystem-driven pathway** (Method 2) as the default rather than the exception – using upstream multi-stakeholder workshops to surface and prioritise challenges across a broader startup pool.

How to ensure long-term diffusion and model adoption?

The AI4SU program structured immediate diffusion (model audit, delivery workshops, open-source publication), but the question of what happens 12 months later remains open. Future programs should consider building in a **follow-up mechanism**: a structured check-in call or short survey with partner startups one year after the hackathon, assessing whether and how models were integrated into operations, what

barriers were encountered, and what additional support would be needed. Platform-level metrics on the AI4SU site (model downloads, notebook views, dataset access) would provide complementary quantitative evidence. Another promising avenue would be to establish **diffusion partnerships with incubators and research centres** – organisations with ongoing relationships with startups and the technical capacity to support model adaptation and deployment beyond the program's own timeline.

How to formalise institutional data partnerships for challenge design?

The Tolbi challenge (crop classification using satellite imagery, in partnership with the BNEDT in Côte d'Ivoire) demonstrated the potential of combining **startup operational needs with institutional data assets**. Research centres, national statistics offices, and public agencies hold large, well-structured datasets that are rarely made accessible for applied AI development. Formalising partnerships with these actors – through data-sharing agreements adapted from the AI4SU framework – could unlock a new category of challenges: ones where the dataset comes from a public or research institution and the startup brings the application layer. This model would reduce the data-sharing burden on startups while creating new pathways for public data to generate economic value through the startup ecosystem.

ANNEXES

Annex 1 – Overview of the 8 AI4SU Hackathons

#	Challenge	Country	Start	End	Modality	Partner	Participants
1	Pixel-Level Crop Classification	Côte d'Ivoire	24 Apr 2025	16 Jun 2025	Online	Tolbi	866
2	Customer Segmentation for Agri-Ecommerce	Ghana	24 Oct 2025	25 Oct 2025	Event	AfriMas h	50
3	Inclusive Credit Scoring for Smallholders	Kenya	29 Oct 2025	31 Oct 2025	Event	FarmSa wa	91
4	Maize Price Intelligence	Kenya	14 Nov 2025	28 Dec 2025	Online	agriBOR A	1 095
5	Produce Recommendation for Food Waste	Kenya	28 Nov 2025	28 Dec 2025	Online	Farm to Feed	733
6	Climate & Market Resilience	Kenya	3 Dec 2025	5 Dec 2025	Event	FarmSa wa	126
7	Predicting Practice Adoption	Kenya	23 Jan 2026	28 Feb 2026	Online	DigiCow	896
8	Soil Nutrient Prediction	Kenya	4 Feb 2026	6 Mar 2026	Online	Rhea	896

Annex 2 – Abbreviations

Abbreviation	Full term
AI	Artificial Intelligence
AI4SU	AI 4 Startups (program)
API	Application Programming Interface
BNEDT	Bureau National d'Études Techniques et de Développement (Côte d'Ivoire)
BP	Best Practice
CTO	Chief Technology Officer
DFI	Development Finance Institution
DSA	Data Sharing Agreement
EU	European Union
FAQ	Frequently Asked Questions
IFC	International Finance Corporation
IP	Intellectual Property
M&E	Monitoring and Evaluation
ML	Machine Learning
MOOC	Massive Open Online Course
MVP	Minimum Viable Product
RMSE	Root Mean Square Error

Annex 3 – References

Program publications

- Digital Africa (2026). *Best Practices for Hackathons with Startups (Focus AI4SU)*. This report.
- Digital Africa (2026). *AI & Data in African Agriculture: Startup Needs and Use Cases Analysis Report*. Companion report to the AI4SU program.
- AI4SU Platform – open-source models, datasets, and documentation: ai4su.digital-africa.co

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External references

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